

Power BI Assignment

Import the provided **JCars Logistics dataset** into your **Aiven PostgreSQL database**. After that, connect your Aiven database to **Power BI** and work on the tasks provided in the file.

As you work, make sure you take clear screenshots of every important step, including:

1. Importing the dataset into your Aiven database
2. Configuring the database connection
3. Adding and using the SSL certificate
4. Connecting Power BI to your Aiven PostgreSQL database

Your article this week will focus on **connecting Power BI to SQL databases**, covering both:

1. A local SQL database
2. A cloud-based SQL database such as Aiven, including SSL certificate configuration

Ensure your screenshots are clear, well-organized, and arranged step by step because they will support your article documentation.

JCars Logistics Company — Power BI Assessment Dataset

Scenario

JCars Logistics imports, sells, and delivers cars to customers across different regions in Kenya. You have been given a raw flat table from the company's sales and logistics records.

Your task is to prepare the data, build a Power BI report, create meaningful calculations, and design an interactive dashboard that can support management decision-making.

Instructions

1. Import the raw flat table into your Aiven PostgreSQL database.
2. Connect the Aiven database to Power BI.
3. Prepare and clean the data so that it can support reliable analysis and dashboard visuals.
4. Create the required calculations and measures needed to answer the business questions.
5. Design an interactive dashboard with clear KPIs, slicers, charts, and a professional layout.
6. Write at least **five insights** and **three recommendations** from your dashboard.

Required Dashboard Questions

Your dashboard should help answer the following business questions:

1. What is the total number of cars sold?

2. What is the total sales revenue recorded by the company?
3. What is the total gross profit and gross profit margin?
4. Which car make generated the highest sales revenue?
5. Which car model had the highest number of units sold?
6. Which vehicle type performed best by revenue and by units sold?
7. Which branch generated the highest revenue?
8. Which region or county had the strongest sales performance?
9. Which sales representative generated the highest revenue and the highest units sold?
10. Which lead source or sales channel produced the most valuable customers?
11. What is the monthly trend of revenue, units sold, and gross profit?
12. Which payment method contributed the most to total revenue?
13. What is the payment status breakdown by revenue and number of orders?
14. Which delivery status has the highest number of orders?
15. What is the average delivery time in days by region, branch, and car make?
16. Which branch has the highest logistics cost compared to sales revenue?
17. Which car make or model gives the best profit margin?
18. Which customer type contributes the highest revenue?
19. What is the relationship between discount and sales revenue?
20. What is the relationship between customer rating and revenue or units sold?
21. Which orders or customers should management investigate further based on the dashboard?
22. Who are the top 10 customers by sales revenue?
23. Which cars are most frequently returned or cancelled?
24. What are the best-performing cars by year of manufacture?
25. What business recommendations would you give JCars Logistics based on your findings?

Minimum Dashboard Requirements

Your Power BI dashboard should include:

- KPI cards for revenue, units sold, number of orders, gross profit, average delivery days, and average rating
- At least one trend chart
- At least one map or regional performance visual
- At least one ranking visual for car make or model performance
- At least one visual comparing revenue, costs, and profit
- Slicers for date/month, region, branch, car make, vehicle type, and payment status
- A short insights section or a summary page

Submission Instructions

Please submit the following:

1. Upload your Power BI file to GitHub and add a clear README file on GitHub explaining:
 - Your project objective

- How you imported the data into Aiven
- How you connected Power BI to the database
- The measures and calculations you created
- The visuals included in your dashboard
- Your key insights
- Your recommendations

submit the GitHub link to datascience@luxdevhq.com by **Sunday, 5th July 2026 4PM**.

2. Publish your article on **Dev.to** explaining how to connect Power BI to SQL databases, both local and cloud-based. Include clear, well-organized screenshots, arranged step by step along with explanations to support your article documentation. Submit your Dev.to article link by **Sunday, 5th July 2026 4PM**

Please ensure your work is neat, well-documented, and professional.